

Prediction Model for Progression-Free Survival in Breast Cancer Patients

P8106 Data Science II — Midterm Project

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Part I: Prediction Model for the 1990–1995 Cohort

1.1 Model Training

Because all 2,000 patients in `datA` were followed until disease progression or death with no censored observations, the log-transformed PFS can be treated as a fully observed continuous outcome. Survival analysis methods (e.g., Cox regression) are designed for censored data and offer no advantage here; standard linear regression-based models are therefore appropriate. The outcome $\log(\text{PFS})$ was approximately normally distributed (see Figure 1), with mean 2.20 (SD = 0.78), further supporting this choice.

All 15 predictor variables were included as candidate predictors, spanning demographics (age, menopausal status, race, BMI), tumor characteristics (tumor size, grade, lymph node count, ER, PR, HER2 status, Ki-67, clinical stage), and initial treatment (surgery type, chemotherapy, hormonal therapy). This decision reflects the primary goal of clinical interpretability rather than parsimony: excluding predictors based on univariate screening or automated selection can introduce bias, and the full set of variables represents established clinical factors for PFS prediction in breast cancer. Regularization within Elastic Net penalizes uninformative predictors implicitly, and the GAM smooth terms adaptively allocate complexity, so over-fitting risk is managed by the modeling framework rather than by variable exclusion. Categorical variables were factor-coded with reference levels set to White race, pre-menopausal status, well-differentiated tumor grade, Stage I, receptor-negative status, breast-conserving surgery, no chemotherapy, and no hormonal therapy.

Exploratory analysis (Figures 2–4) identified two features motivating the candidate model set. First, tumor size and lymph node count were strongly collinear (Pearson $r = 0.71$), inflating

variance under OLS and motivating penalized regression. Second, LOESS curves revealed non-linear associations of log(PFS) with BMI, tumor size, and lymph node count, motivating flexible additive models.

Four candidate models were trained in R using the caret package (v6.0) with `set.seed(1)` applied before every stochastic operation. Model evaluation used 10-fold cross-validation repeated 5 times (50 resamples total), specified via `trainControl(method = "repeatedcv", number = 10, repeats = 5)`. Predictors were centered and scaled using `preProcess = c("center", "scale")` applied inside each CV fold (not on the full training set before splitting), ensuring no data leakage into the validation folds. The same 50-resample index was used across all caret models for paired comparison via `resamples()`.

Model 1 — OLS fit a standard multiple linear regression with all 16 predictors, serving as the interpretable baseline.

Model 2 — Elastic Net combined L1 and L2 penalties via the objective $\min \|y - X\beta\|^2 + \lambda[(1-\alpha)\|\beta\|^2 + \alpha\|\beta\|_1]$. The mixing parameter α was searched over $\{0, 0.1, \dots, 1\}$ and λ over 50 log-uniformly spaced values in $[0.001, 1]$. Optimal hyperparameters were selected by minimizing CV-RMSE.

Model 3 — MARS searched over degree $\in \{1, 2\}$ and number of retained terms $\in \{2, \dots, 20\}$. The optimal model selected degree = 1, indicating no two-way interaction improved out-of-sample performance.

Model 4 — GAM was fitted using `mgcv::gam()` with REML smoothness estimation. Penalized thin-plate regression splines were placed on each continuous predictor; categorical predictors entered linearly. CV-RMSE for the GAM was obtained via a single 10-fold split (not repeated) using the same seed.

1.2 Results

As shown in Table 3, all four models yielded nearly identical CV-RMSE values (range: 0.6975–0.7035). The absolute differences between models are small (< 0.006 RMSE units) and likely not practically significant given the inherent predictive ceiling of 16 baseline clinical variables. GAM was selected as the final model for the following reasons: (1) it achieved the lowest CV-RMSE

(0.6975) and highest CV-R² (0.2068); (2) the LOESS-identified non-linearities provided empirical justification for smooth terms; and (3) MARS selecting degree = 1 confirmed that no interaction term improved prediction, validating the additive structure. The final selected model is a Generalized Additive Model of the form:

$$\log(\text{PFS})_i = \beta_0 + s(\text{age}) + s(\text{BMI}) + s(\text{tumor_size}) + s(\text{lymph_nodes}) + s(\text{Ki67}) + \mathbf{X}_i^T \boldsymbol{\gamma} + \varepsilon_i$$

where $s(\cdot)$ denotes a penalized thin-plate regression spline estimated by REML, $\mathbf{X}_i$ is the vector of categorical predictors entering linearly with coefficient vector $\boldsymbol{\gamma}$, and $\varepsilon_i \sim N(0, \sigma^2)$ is the error term. The effective degrees of freedom for each smooth term are determined by the data.

The parametric coefficient estimates are presented in Table 3a, and smooth-term significance is summarized in Table 3b. Significant predictors included: ER-positive status (+0.234 log-months, $p < 0.001$), HER2-positive status (−0.174, $p < 0.001$), poorly differentiated grade (−0.162, $p = 0.006$), and hormonal therapy (+0.133, $p < 0.001$). All smooth terms were significant, with BMI showing the strongest non-linearity (edf = 4.00; Table 3b). Partial effects are shown in Figure 5.

The final GAM had a training RMSE of 0.690 and training adjusted R² of 0.214; cross-validated RMSE was 0.698 and CV-R² was 0.207 (Table 3c). The negligible gap between training and CV metrics confirms that overfitting was not a concern.

Part II: Generalizability to the 2000–2005 Cohort

2.1 Cohort Comparison

The 2000–2005 cohort (datB) differed substantially from datA on both outcome and predictors (Tables 4–6). Mean log(PFS) increased from 2.196 to 2.742 (Welch t-test, $p < 2.2 \times 10^{-16}$; Figure 6), consistent with advances in treatment, earlier detection, and changing screening practices. The datB cohort showed a higher proportion of Stage I disease (34.6% vs. 25.2%), greater chemotherapy (64.9% vs. 55.2%) and hormonal therapy use (57.0% vs. 46.7%), and lower mean tumor size and lymph node count.

2.2 External Validation of Model A

The datA GAM was applied directly to datB using `predict()` with no model updating. Generalizability was assessed against two practical heuristic criteria: external RMSE $\leq 110\%$ of internal CV-RMSE (allowing at most a 10% degradation in precision) and external $R^2 \geq 90\%$ of internal CV- R^2 (allowing at most a 10% degradation in explained variance). These thresholds are not derived from formal theory but represent reasonable benchmarks for applied clinical prediction models, balancing the expectation of some performance loss on a new population against the requirement for a usable model.

Both criteria failed (Table 8). External RMSE was 0.844 (vs. CV-RMSE of 0.698, a 21% increase) and external R^2 was 0.162 (vs. 0.207, a 22% decrease; Tables 7–8). The mean prediction bias was +0.500, meaning the model systematically underpredicted log(PFS) in datB by approximately half a log-month. The quintile calibration table (Table 8a) and calibration plot (Figure 7) reveal that this error has the structure of a systematic intercept shift: observed log(PFS) exceeded predictions by a nearly constant offset (~ 0.50) across all five quintiles of predicted values, with the slope of observed-on-predicted being approximately correct (Figure 8). This pattern indicates that the model's relative risk ordering and association structure are preserved, but the absolute level of PFS has shifted upward in the later cohort—a distributional shift rather than full model misspecification. The model therefore does not generalize in absolute terms, but the association structure remains informative.

2.3 Refitted Model in datB

The same four-model training pipeline was applied to datB with identical preprocessing, seed, and resampling. As shown in Table 9, GAM again achieved the best performance (CV-RMSE = 0.669, CV- R^2 = 0.177), confirming that the nonlinear additive structure is the appropriate model class for this outcome. The refitted GAM improved substantially over the transported model: RMSE fell from 0.844 to 0.669 and MAE from 0.686 to 0.534 (Table 10).

2.4 What Changed in Prediction

Comparison of parametric coefficients and smooth-term significance between the two GAMs reveals meaningful shifts (Tables 11–12). Most notably, HER2 status reversed: in datA it was

significantly associated with worse PFS (estimate = -0.174 , $p < 0.001$); in datB the coefficient was near zero and non-significant (estimate = $+0.034$, $p = 0.35$), likely reflecting wider access to HER2-directed therapy in the early 2000s. Hormonal therapy's benefit strengthened (datA: $+0.133$; datB: $+0.209$, both $p < 0.001$). Conversely, the effect of poorly differentiated grade attenuated (datA: -0.162 , $p = 0.006$; datB: -0.089 , $p = 0.13$) and age became non-significant in datB ($p = 0.10$). Among smooth terms, Ki-67 gained significance (datB $p = 0.007$ vs. 0.050 in datA) and lymph node count showed greater non-linearity (edf: $1.00 \rightarrow 1.69$; Figure 9).

In summary, the best model class, GAM with smooth terms on continuous predictors, remained consistent across cohorts. However, the fitted prediction function shifted materially, primarily driven by the population-level increase in PFS and the changing predictive role of HER2. The systematic underprediction in datB reflects an intercept shift rather than structural misspecification; re-estimating the model in datB fully corrected this shift and produced materially better predictions.

Appendix: Tables and Figures

Table 1. Summary statistics of continuous variables in datA (1990–1995).

Variable	Mean	SD	Median	Min	Max
Age (years)	54.99	11.65	55.00	25.00	85.0
BMI (kg/m ²)	26.06	4.56	26.05	16.00	42.6
Tumor size (cm)	3.38	1.71	3.10	0.30	8.3
Lymph nodes (count)	2.62	2.59	2.00	0.00	16.0
Ki-67 (%)	21.85	11.58	21.60	1.00	62.3
log(PFS) (months)	2.20	0.78	2.19	−0.53	5.1

Table 2. Frequency distribution of categorical variables in datA (1990–1995).

Variable	Level	n (%)
Menopausal status	Pre-menopausal	689 (34.5%)
	Post-menopausal	1311 (65.5%)
Race/ethnicity	White	1284 (64.2%)
	Black	289 (14.4%)
	Asian	176 (8.8%)
	Hispanic	251 (12.6%)
Tumor grade	Well differentiated	383 (19.1%)
	Moderately differentiated	871 (43.5%)
	Poorly differentiated	746 (37.3%)
Clinical stage	Stage I	504 (25.2%)
	Stage II	893 (44.6%)
	Stage III	603 (30.1%)
ER status	Positive	1281 (64.0%)
PR status	Positive	961 (48.0%)
HER2 status	Positive	444 (22.2%)
Surgery type	Breast-conserving	986 (49.3%)
	Mastectomy	1014 (50.7%)
Chemotherapy	Yes	1105 (55.2%)
Hormonal therapy	Yes	934 (46.7%)

Table 3. Cross-validated performance of four candidate models on datA (10-fold CV $\times$ 5 repeats). Performance differences are small (< 0.006 RMSE) and not practically significant; GAM was selected for its lowest CV-RMSE and empirically justified non-linearity.

Model	CV-RMSE	CV-R ²
GAM (selected)	0.6975	0.2068
Elastic Net	0.7016	0.1996
MARS	0.7031	0.1979
OLS	0.7035	0.1951

Table 3a. Parametric coefficient estimates from the final GAM (datA). Reference levels: White, pre-menopausal, well-differentiated, Stage I, receptor-negative, BCS, no chemotherapy, no hormonal therapy.

Term	Estimate	SE	t-value	p-value
(Intercept)	2.013	0.078	25.98	<0.001
Age	(smooth — Table 3b)			0.029
Menopausal: Post	0.038	0.047	0.81	0.416
Race: Black	0.038	0.045	0.84	0.400
Race: Asian	0.067	0.056	1.19	0.234
Race: Hispanic	−0.042	0.048	−0.87	0.386
BMI	(smooth — Table 3b)			<0.001
Tumor size	(smooth — Table 3b)			0.006
Grade: Moderate	−0.045	0.046	−0.97	0.332
Grade: Poor	−0.162	0.059	−2.74	0.006
Lymph nodes	(smooth — Table 3b)			<0.001
ER: Positive	0.234	0.052	4.53	<0.001
PR: Positive	−0.006	0.039	−0.14	0.885
HER2: Positive	−0.174	0.039	−4.49	<0.001
Ki-67	(smooth — Table 3b)			0.050
Stage: II	0.085	0.056	1.52	0.128
Stage: III	0.095	0.104	0.91	0.364
Surgery: Mastectomy	−0.008	0.033	−0.25	0.803
Chemotherapy: Yes	−0.001	0.032	−0.04	0.965
Hormonal therapy: Yes	0.133	0.040	3.33	<0.001

Table 3b. Smooth-term significance in the final GAM (datA). edf > 1 indicates non-linearity.

Smooth Term	edf	Ref. df	F	p-value
s(Age)	1.001	1.001	4.74	0.029
s(BMI)	3.995	4.988	7.07	<0.001

Smooth Term	edf	Ref. df	F	p-value
s(Tumor size)	2.121	2.729	4.55	0.006
s(Lymph nodes)	1.002	1.004	46.35	<0.001
s(Ki-67)	1.001	1.002	3.83	0.050

Table 3c. Performance metrics for the final GAM (datA). The negligible train–CV gap indicates no overfitting.

Metric	Value
Training RMSE	0.6899
10-fold CV-RMSE (5 repeats)	0.6975
Training R ² (adjusted)	0.2145
CV-R ²	0.2068

Table 4. Comparison of log(PFS) between cohorts (Welch t-test, $p < 2.2 \times 10^{-16}$).

Cohort	n	Mean log(PFS)	SD	Median
datA (1990–1995)	2000	2.196	0.783	2.191
datB (2000–2005)	2000	2.742	0.738	2.744

Table 5. Continuous predictor comparison between cohorts (Welch t-test).

Variable	datA Mean (SD)	datB Mean (SD)	Diff (B–A)	p-value
Age (years)	54.99 (11.65)	56.05 (11.22)	+1.06	0.004
BMI (kg/m ²)	26.06 (4.56)	27.44 (5.05)	+1.38	<0.001
Tumor size (cm)	3.38 (1.71)	3.08 (1.68)	–0.31	<0.001
Lymph nodes	2.62 (2.59)	2.32 (2.53)	–0.31	<0.001
Ki-67 (%)	21.85 (11.58)	20.37 (11.71)	–1.48	<0.001

Table 6. Treatment and stage distribution between cohorts.

Variable	datA (%)	datB (%)
Stage I / II / III	25.2 / 44.6 / 30.1	34.6 / 41.4 / 24.0
Chemotherapy (Yes)	55.2	64.9
Hormonal therapy (Yes)	46.7	57.0
Mastectomy	50.7	44.1

Table 7. External validation performance of the datA GAM on datB versus internal CV performance. The large positive bias (+0.500) reflects a systematic intercept shift rather than structural misspecification.

Setting	RMSE	MAE	R ²	Mean Bias
Internal CV on datA	0.6975	—	0.2068	—
External validation on datB	0.8436	0.6862	0.1617	+0.5003

Table 8. Generalizability criteria applied to the datA GAM. Thresholds ($\leq 110\%$ RMSE degradation; $\geq 90\%$ R² retention) represent practical heuristics, not formal bounds.

Criterion	Threshold	Observed	Result
External RMSE $\leq 110\%$ of internal CV-RMSE	≤ 0.7673	0.8436	Fail
External R ² $\geq 90\%$ of internal CV-R ²	≥ 0.1861	0.1617	Fail
Overall conclusion	—	—	Does not generalize

Table 8a. Quintile-based calibration of the datA GAM in datB. The nearly constant gap (~ 0.64) across quintiles confirms an intercept shift rather than slope distortion.

Quintile of predicted value	Mean predicted log(PFS)	Mean observed log(PFS)	Difference	n
1 (lowest)	1.691	2.329	+0.638	400
2	2.051	2.544	+0.493	400
3	2.275	2.758	+0.483	400
4	2.480	2.951	+0.471	400
5 (highest)	2.713	3.131	+0.418	400

Table 9. Cross-validated model performance on datB (10-fold CV $\times$ 5 repeats). GAM again achieves the lowest CV-RMSE; differences across models remain small.

Model	CV-RMSE	CV-R ²
GAM (selected)	0.6693	0.1772
Elastic Net	0.6726	—
MARS	0.6731	0.1707
OLS	0.6735	0.1691

Table 10. Performance of the transported datA GAM versus the refitted datB GAM.

Model	RMSE	MAE	R ²
Transported GAM from datA	0.8436	0.6862	0.1617
Refitted GAM on datB	0.6693	0.5337	0.1772

Table 11. Parametric GAM coefficients: datA vs. datB. Asterisk (*) denotes $p < 0.05$. The HER2 reversal and strengthening of hormonal therapy are the most clinically meaningful changes.

Term	Est. (datA)	p (datA)	Est. (datB)	p (datB)
(Intercept)	2.013	<0.001*	2.520	<0.001*
Menopausal: Post	0.038	0.416	-0.034	0.446
Race: Black	0.038	0.400	-0.040	0.319
Race: Asian	0.067	0.234	-0.039	0.427
Race: Hispanic	-0.042	0.386	0.094	0.027*
Grade: Moderate	-0.045	0.332	0.079	0.063
Grade: Poor	-0.162	0.006*	-0.089	0.126
Stage: II	0.085	0.128	-0.004	0.934
Stage: III	0.095	0.364	0.026	0.807
ER: Positive	0.234	<0.001*	0.142	0.012*
PR: Positive	-0.006	0.885	0.064	0.081
HER2: Positive	-0.174	<0.001*	0.034	0.350
Surgery: Mastectomy	-0.008	0.803	-0.017	0.602
Chemotherapy: Yes	-0.001	0.965	-0.020	0.559
Hormonal therapy: Yes	0.133	<0.001*	0.209	<0.001*

Table 12. Smooth-term comparison between datA and datB GAMs. s(Ki-67) gained significance and s(lymph_nodes) became more non-linear in datB.

Smooth Term	edf (datA)	p (datA)	edf (datB)	p (datB)
s(Age)	1.001	0.029*	1.001	0.103
s(BMI)	3.995	<0.001*	3.645	<0.001*
s(Tumor size)	2.121	0.006*	2.142	0.010*
s(Lymph nodes)	1.002	<0.001*	1.691	0.005*
s(Ki-67)	1.001	0.051	1.001	0.007*

Figures

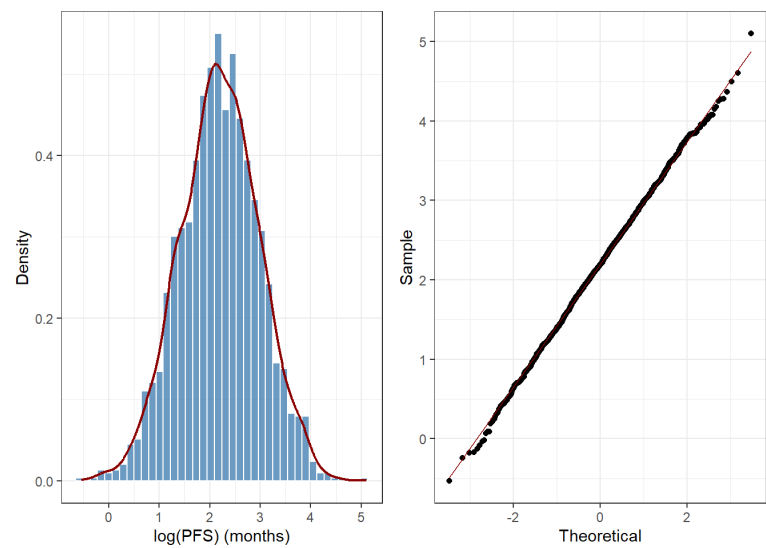


Figure 1. Distribution of log(PFS) (histogram + density overlay) and normal Q-Q plot for dataA. The approximately normal distribution supports linear regression-based modeling without survival analysis.

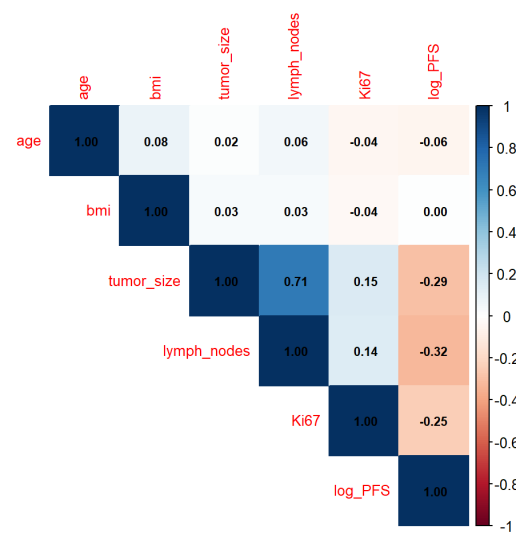


Figure 2. Correlation matrix of continuous predictors in dataA. Tumor size and lymph node count are strongly collinear ($r = 0.71$), motivating Elastic Net regularization.

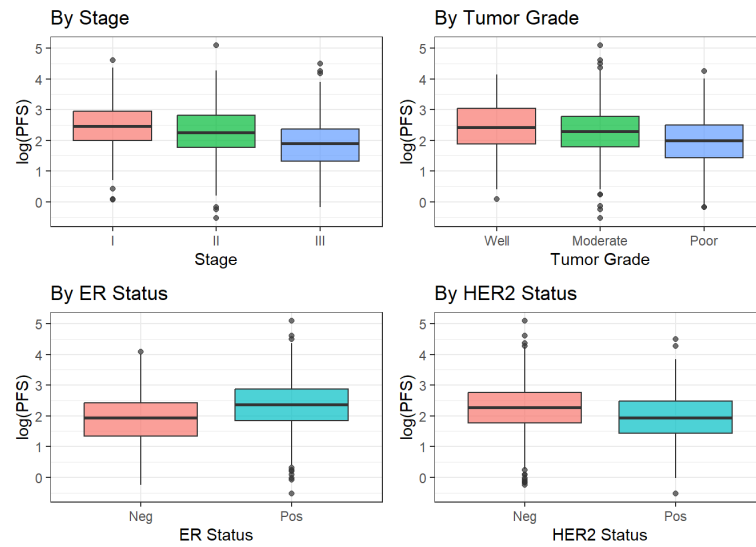


Figure 3. Boxplots of $\log(\text{PFS})$ by clinical stage, tumor grade, ER status, and HER2 status in datA.

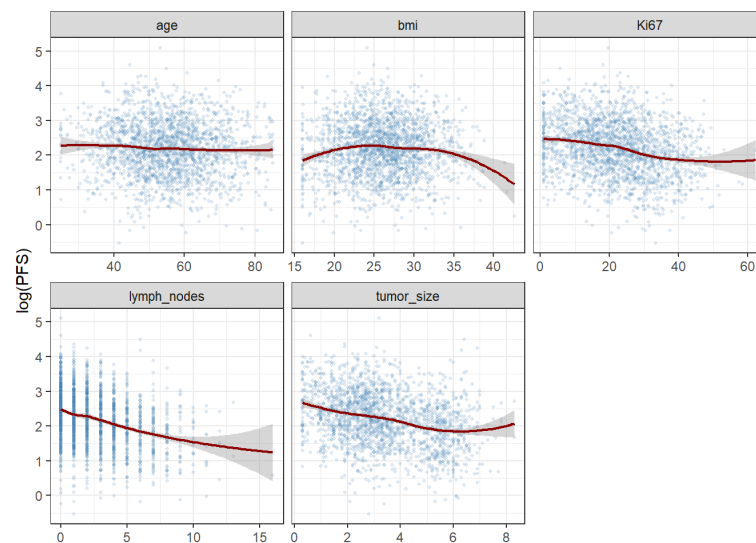


Figure 4. LOESS scatterplots of $\log(\text{PFS})$ against each continuous predictor in datA. Non-linear trends are evident for BMI, tumor size, and lymph node count.

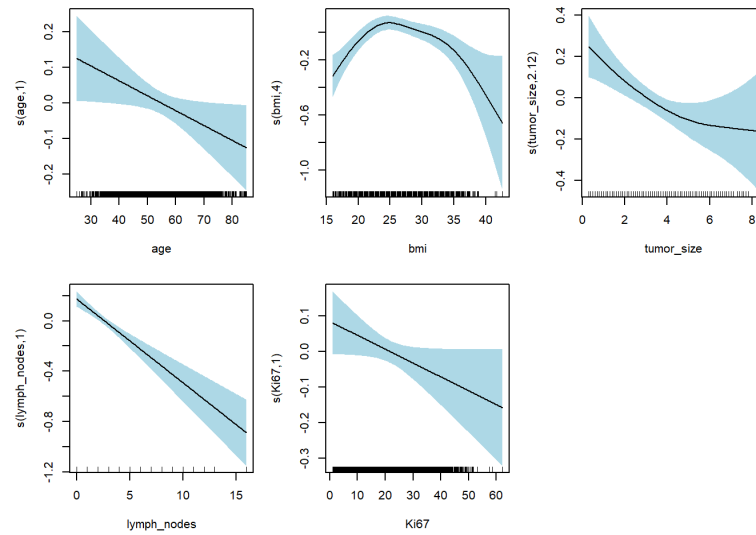


Figure 5. Partial effects of smooth terms in the final datA GAM. Shaded bands represent 95% credible intervals; rug marks indicate data density.

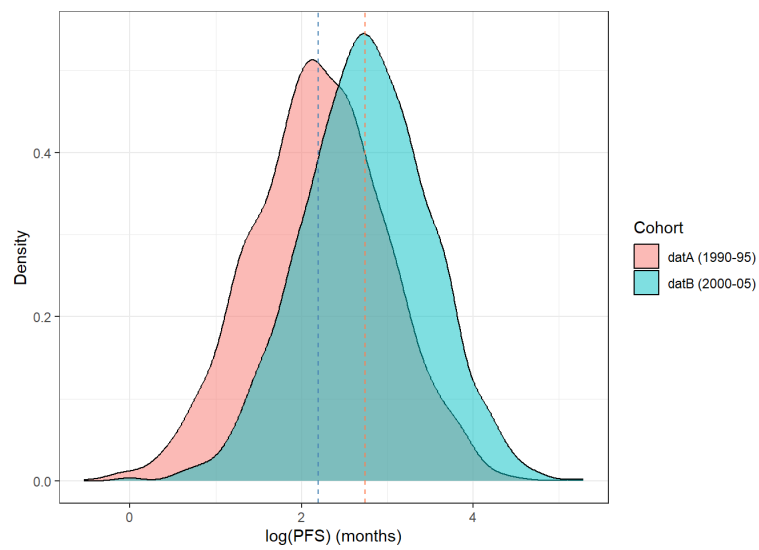


Figure 6. Density of log(PFS) in datA (1990–1995) versus datB (2000–2005). The rightward shift reflects improvements in treatment outcomes over the decade.

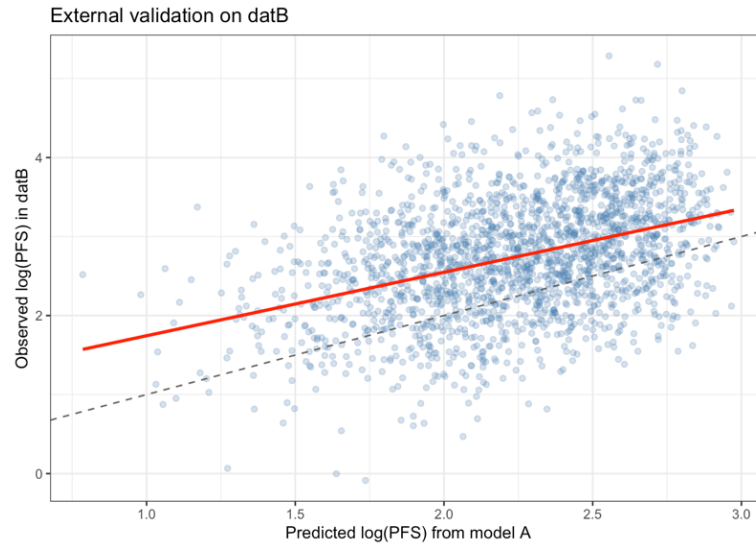


Figure 7. Quintile-based calibration plot: datA GAM applied to datB. The near-constant vertical gap (~ 0.50 log-months) across all quintiles confirms a systematic intercept shift rather than slope distortion.

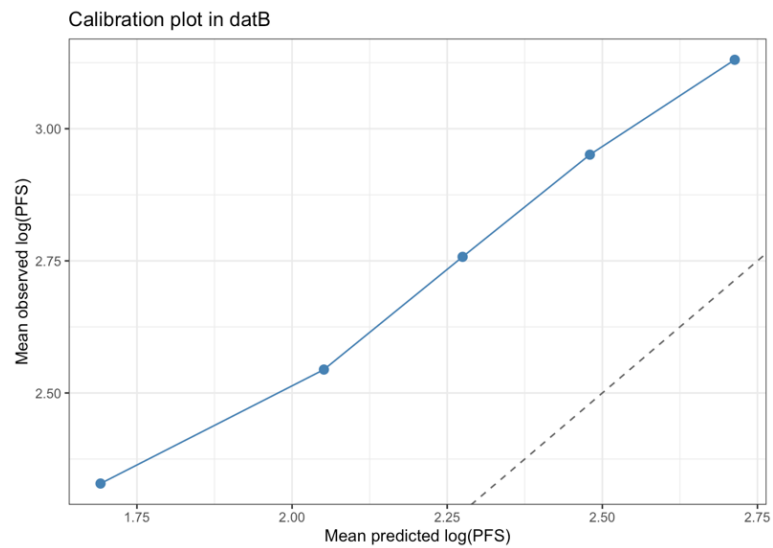


Figure 8. Observed-vs-predicted log(PFS) in datB using the datA GAM. The fitted regression line (red) lies below the 45° reference (dashed), confirming systematic underprediction consistent with an intercept shift.

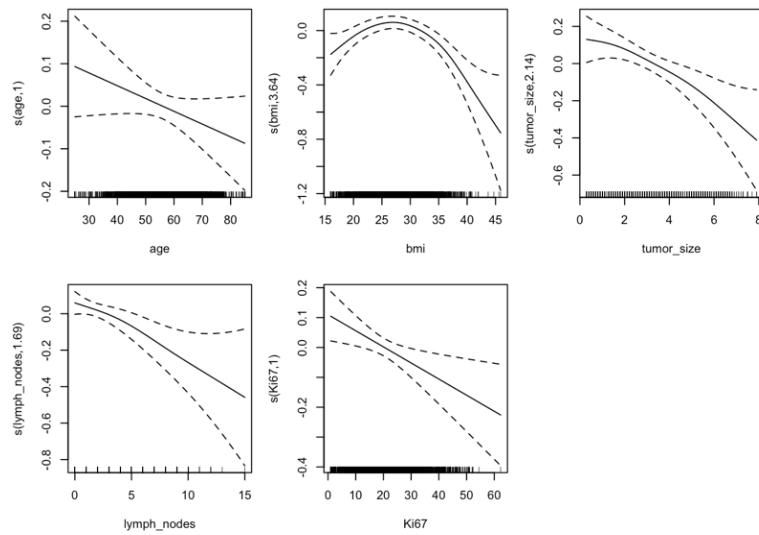


Figure 9. Partial effects of smooth terms in the refitted datB GAM. Compared with Figure 5, $s(\text{lymph_nodes})$ shows greater non-linearity (edf = 1.69 vs. 1.00) and $s(\text{Ki-67})$ displays a clearer negative linear trend.